1 Appendix for Section 4

1.1 Apparent Balanced Growth in $\bar{c}$ and $\text{cov}(c, \boldsymbol{p})$

{sec:ApndxBalance

Section 4.2 demonstrates some propositions under the assumption that, when an economy satisfies the GIC, there will be constant growth factors $\Omega_{\bar{c}}$ and Ω_{cov} respectively for $\bar{c}$ (the average value of the consumption ratio) and $\text{cov}(c, \boldsymbol{p})$. In the case of a Szeidl-invariant economy, the main text shows that these are $\Omega_{\bar{c}} = 1$ and $\Omega_{\text{cov}} = \mathcal{G}$. If the economy is Harmenberg- but not Szeidl-invariant, no proof is offered that these growth factors will be constant.

1.2 $\log c$ and $\log \text{cov}(c, \boldsymbol{p})$ Grow Linearly

Figures 1 and 2 plot the results of simulations of an economy that satisfies Harmenberg- but not Szeidl-invariance with a population of 4 million agents over the last 1000 periods (of a 2000 period simulation).[1] The first figure shows that $\log \bar{c}$ increases apparently linearly. The second figure shows that $\log(-\text{cov}(c, \boldsymbol{p}))$ also increases apparently linearly. (These results are produced by the notebook `ApndxBalancedGrowthcNrmAndCov.ipynb`).

[1] For an exposition of our implementation of Harmenberg's method, see this supplemental appendix.

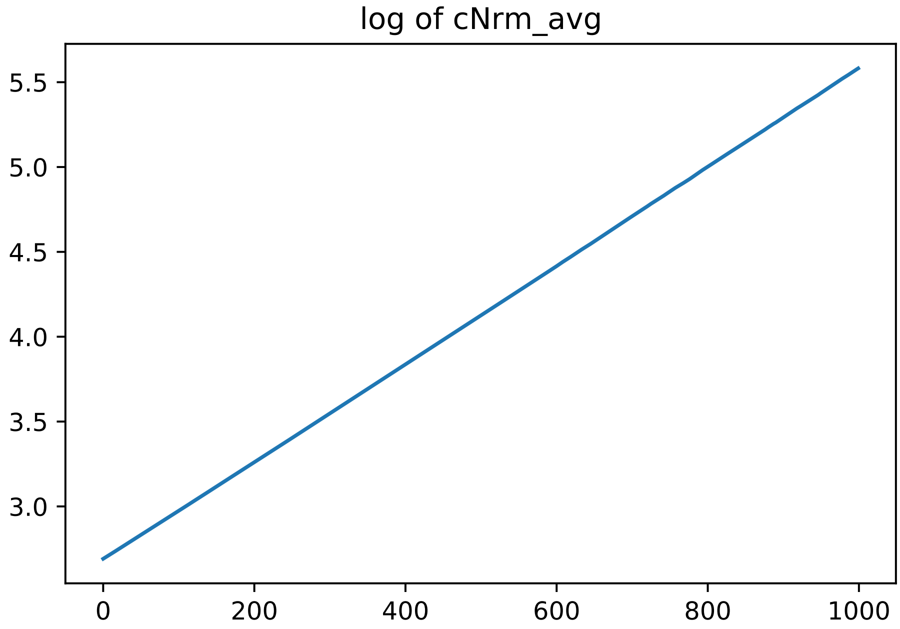

Figure 1 Appendix: $\log\,\mathfrak{c}$ Appears to Grow Linearly {fig:logcNrm}

(see (2))

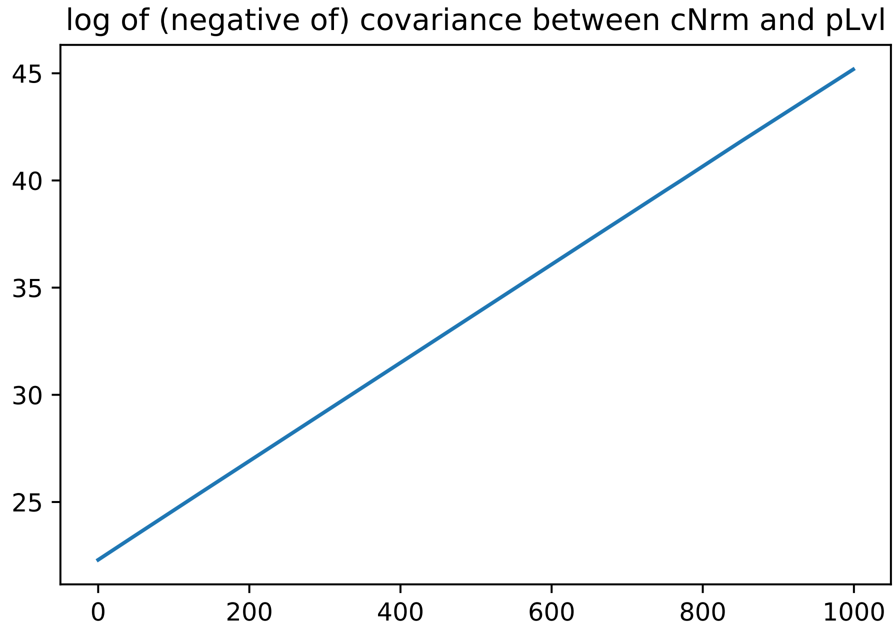

Figure 2 Appendix: $\log\left(-\operatorname{cov}(c, \boldsymbol{p})\right)$ Appears to Grow Linearly {fig:logcov}